Bilateral Real Exchange Rates and Korean Firms' Export and Import Activities

Evidence from Trade by Enterprise Characteristics

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Abstract

This paper examines Korean firms' export and import responses to movements in bilateral real exchange rates between Korea and its trading partner countries. We provide the first firm–country–product-level analysis of these responses in Korea, using newly constructed Trade by Enterprise Characteristics (TEC) data for 2015–2024. A 1% real depreciation of the Korean won is associated with increases of 0.229% in export values and 0.512% in import values, both measured in won. Estimated export responses are larger among smaller firms. Exporters expand their product range and reduce product concentration within existing destinations, with no significant changes in market entry or continuation. Importers exhibit greater product concentration within source countries and are more likely to end existing sourcing relationships. At the firm level, greater import-side exposure to depreciation is associated with lower employment, higher total sales, and more concentrated sourcing across countries, whereas export-side exposure is only weakly associated with higher sales.

Keywords: bilateral real exchange rates; Trade by Enterprise Characteristics; exports and imports; firm heterogeneity; market entry and continuation

JEL classification: F14, F31, L25

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1 Introduction

Exchange rate fluctuations can have substantial effects on the Korean economy, given its high dependence on international trade and integration into global supply chains. By affecting export revenues and the cost of imported intermediate inputs and energy, movements in the Korean won (KRW) influence corporate profitability, domestic prices, and economic activity, with broader implications for macroeconomic stability.

Despite this importance, access to and use of Korean microdata that combine firm identifiers with trading-partner and product information and link these records to firm characteristics have been limited. Existing firm-level studies have therefore generally examined exchange rates and firm activity using total exports and imports aggregated across trading partners. Analyses focusing on the won–dollar exchange rate face two limitations in particular.

First, the won–dollar exchange rate alone cannot adequately capture the exchange rate changes firms face in their transactions with individual trading partners. As Figure 1 shows, the won's nominal exchange rates against major currencies do not always move in the same direction. For example, over 2022–2024, the won price of the dollar generally increased, while the won price of 100 yen trended downward. Thus, the won can depreciate against the dollar while appreciating against the yen over the same period. Relying solely on the won–dollar exchange rate makes it difficult to distinguish partner-specific exchange rate movements and the associated adjustments in firms' trade.

Second, in an analysis restricted to Korean firms, the won–dollar exchange rate is common to all firms at a given point in time. Its coefficient cannot therefore be estimated separately when time fixed effects are included. Omitting time fixed effects, in turn, may cause the exchange rate coefficient to capture common factors, such as global economic or financial conditions, that move with exchange rates. Existing studies address this issue by using industry-specific real effective exchange rates (Han and Pyun, 2021) or interactions between the won–dollar exchange rate and firm size (Kwon et al., 2025), allowing time fixed effects to be retained. These approaches are useful for identifying industry exposure or differences in responses across firm sizes, but they do not separately reveal how individual firms respond to exchange rate movements that differ across their trading partners.

This study combines firm–country–product transactions from Trade by Enterprise Characteristics (TEC), jointly compiled by Statistics Korea and the Korea Customs Service, with bilateral real exchange rates between Korea and its trading partner countries. The main sample covers 2015–2024, when firm-size information is available. The analysis of firm-level outcomes, including employment and sales, uses establishments linked to the Mining and Manufacturing Survey (MMS). Because bilateral real exchange rate movements differ across countries even within a given year, we can estimate exchange rate

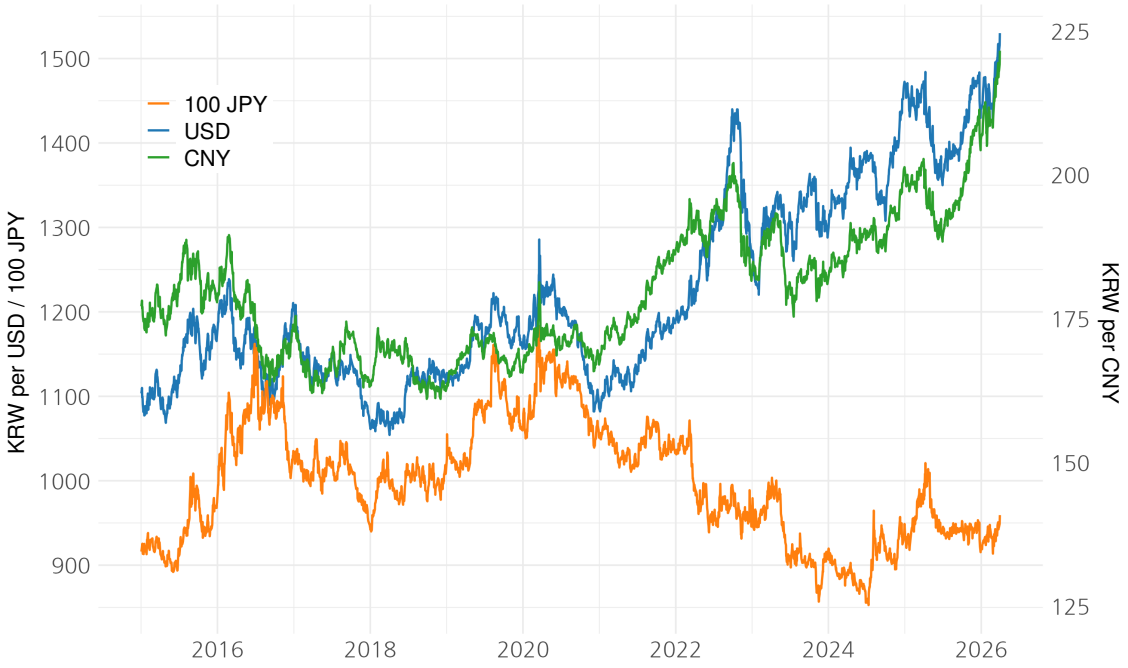


Figure 1. Nominal Exchange Rates of the Korean Won against Major Currencies

Notes: The left axis shows Korean won (KRW) per U.S. dollar and per 100 Japanese yen; the right axis shows KRW per Chinese yuan. An increase denotes a depreciation of the won against the relevant currency. This figure presents bilateral nominal exchange rates as background; the empirical analysis uses bilateral real exchange rates.

Source: Bank of Korea interim seminar presentation, September 9, 2026, slide 2.

coefficients for transactions with particular partners while controlling for common time shocks. We first examine trade values at the firm–country–product level and annual changes in export sensitivity. We then analyze product counts, concentration, market entry, and continuation at the firm–country level. Finally, we construct firm-specific exchange rate exposure using the previous year's trading structure and relate it to total employment, sales, the number of trading partners, and country concentration. Applying the same analytical framework to exports and imports allows us to compare adjustment in sales and sourcing markets.

The study makes three contributions. First, it distinguishes Korean firms' transactions by trading partner and analyzes export and import responses to bilateral real exchange rate movements while controlling for common time shocks. This makes it possible to examine adjustments in transactions with individual countries that are difficult to identify using total firm-level trade. Second, it examines adjustment through trade values, product composition, and market entry and continuation. In particular, we extend the export framework of Li et al. (2015) to imports to compare responses in sales and sourcing markets. Third, we combine bilateral real exchange rate movements with firms' trading structures to construct firm-specific exchange rate exposure and examine its relationship with overall outcomes such as employment and sales.

The main findings are as follows. A 1% real depreciation of the won is associated with an approximately 0.229% increase in won-denominated export values at the firm–country–product level. The response is larger for smaller firms: the coefficients are 0.207 for large firms, 0.224 for mid-sized firms, and 0.238 for small and medium-sized enterprises (SMEs). These size patterns vary across regions. In East Asia, for example, the response of SMEs is notably larger than those of the other two groups and statistically significant, whereas the response of large firms is not significant. Within trading relationships, the number of export products increases and concentration falls in the same destination. By contrast, we find no significant exchange rate response in export market entry or continuation, nor significant annual changes in exchange rate sensitivity. At the firm level, export exchange rate exposure is only weakly associated with sales growth and is not significantly related to changes in employment or the number of trading partners.

These findings suggest that exporters' responses are more apparent in export values and product composition within existing relationships than in their partner-country composition or employment. This pattern is consistent with an interpretation in which firms perceive exchange rate changes as temporary shocks: fixed and adjustment costs associated with market development and hiring or workforce adjustment may limit short-run changes in trading partners and employment.

For imports, a 1% real depreciation of the won is associated with an approximately 0.512% increase in won-denominated import values at the firm–country–product level. This increase also incorporates higher import prices when converted into won and cannot be interpreted directly as an increase in import quantities. Unlike exports, imports become more concentrated across products within the same origin country, without a significant change in the number of products. The probability of discontinuing transactions with an existing origin country also increases, while entry into a new origin country does not change significantly. At the firm level, import exchange rate exposure is associated with lower employment, greater concentration across origin countries, and higher total sales.

Importers' responses thus appear not only in trade values but also in concentration across sourcing products and countries, the termination of existing trading relationships, and employment adjustment. These findings are consistent with the possibility that higher import costs following a depreciation of the won lead firms to discontinue transactions with some origin countries or adjust their sourcing structures and employment.

Trade values alone, however, cannot distinguish price from quantity responses and also incorporate currency conversion effects. Increases in export and import values cannot therefore be equated with an expansion of real trade. Similarly, higher total sales do not necessarily imply an expansion of real firm activity, because price increases cannot be separated from sales-volume growth. Section 7 dis-

cusses these data limitations, together with the possibility of asymmetric responses to appreciation and depreciation of the won.

The remainder of the paper is organized as follows. Section 2 reviews the literature and analytical perspectives. Section 3 describes the data and units of analysis. Section 4 presents the empirical models and sample construction. Sections 5 and 6 discuss the export and import results, respectively. Section 7 brings the findings together and discusses policy implications and limitations.

2 Related Literature and Analytical Perspectives

2.1 Dollar Pricing and the Role of Exchange Rates

Recent work in international finance and macroeconomics has emphasized how the currency in which international transactions are priced affects the role of exchange rates in trade adjustment. The dominant currency paradigm (DCP) of Gopinath et al. (2020) highlights pricing in a dominant currency, such as the U.S. dollar, rather than in either trading partner's currency. When dollar prices are sticky in the short run, the importing country's exchange rate against the dollar determines the prices paid by importers. Bilateral exchange rates between trading partners alone may therefore be insufficient to explain short-run trade-price and quantity responses.

Dollar pricing does not, however, eliminate the role of bilateral exchange rates. Using Chilean export data, De Gregorio et al. (2024) find that the short-run price effect of bilateral exchange rates disappears after controlling for exchange rates against the U.S. dollar, but bilateral exchange rates become more important for export-price and quantity adjustment over time. The roles of exchange rates against the U.S. dollar and bilateral exchange rates between trading partners should therefore be considered jointly, taking account of price rigidity and the adjustment horizon.

2.2 Microeconomic Evidence on Bilateral Exchange Rates and Firm Responses

Alongside the recent discussion of the DCP, a long-standing literature uses firm-level microdata to examine bilateral exchange rates and firms' trade responses. This literature emphasizes differences in responses by productivity, size, import dependence, and product characteristics. Berman et al. (2012) show that more productive firms adjust prices and markups more, and quantities less, in response to exchange rate movements. Amiti et al. (2014) identify imported-input use and market share as important determinants of exchange rate pass-through. Chen and Juvenal (2016) examine differences in price and quantity responses by product quality, while Devereux et al. (2017) analyze the roles of importers' as well as exporters' market shares and currency choices in pass-through. These findings

indicate that differences across firm sizes should be interpreted in conjunction with cost structures, product composition, and trading-partner characteristics.

Bilateral exchange rates between trading partners may affect not only the prices and values of existing transactions but also the formation and continuation of trading relationships and their product composition. Li et al. (2015) use Chinese firm–product–destination data to analyze exchange rates, export prices and quantities, and export market entry and continuation. Their analysis is confined to exports. We extend their approach to imports to compare firms' adjustment to exchange rate movements in sales and sourcing markets. Using Canadian import data, Devereux et al. (2025) show that the formation and termination of relationships between foreign exporters and domestic importers, as well as product adjustment within existing relationships, respond to exchange rates, with larger responses among large firms. Their firm-to-firm relationships differ from our firm–country unit of analysis, but their findings suggest that the ranking of responses by firm size need not be the same for trade values and trading relationships.

Adjustment to exchange rate shocks may also unfold over time. Eaton et al. (2025) explain the gradual accumulation of export responses in a model with customer search costs and learning about foreign demand. Their framework offers a perspective on why short-run trade-value responses may differ from adjustments in market composition. The presence of search or fixed costs alone does not, however, imply that exchange rates have no effect on market entry or exit. Whether and how quickly trading relationships adjust may depend on the size and persistence of shocks and on firm characteristics.

The literature has also extended the analysis to firm-wide exchange rate exposure and real outcomes. Dai and Xu (2017) examine employment responses using exposure measures that reflect firms' export and import partner composition and trade dependence. They show that differences in trading-partner composition are important for understanding employment reallocation across firms. This approach is closely related to our use of bilateral real exchange rate movements and lagged trading structures to examine employment and sales. Blaum (2024) emphasizes that exporters can also be intensive importers and explains how export expansion and imported-input adjustment are jointly determined following a large depreciation. Understanding firm-wide responses therefore requires attention to both export-market and import-sourcing exposure. We apply the same analytical structure to exports and imports to compare these responses.

2.3 Data and Estimation Constraints in Korea and Our Contribution

In Korea, access to and use of data linking firm identifiers, trading partners, and product-level transactions to firm characteristics have been limited. Using product–destination data, Son (2023) examines

invoicing-currency choice and exchange rate pass-through. The study shows that, given the high share of dollar invoicing in Korean exports, prices in importing countries' local currencies respond strongly to exchange rates between the dollar and the importing country's currency. These data are at the product–destination–year level and contain no firm identifiers, however, so they do not permit direct analysis of individual firms' responses.

Firm-level studies have generally examined exchange rates and firm activity using total export or import values aggregated across trading partners. Lee and Naknoi (2024) study deflated total exports across all destinations and export participation. They use industry-specific exchange rates that reflect invoicing currencies to examine price-competitiveness and imported-input cost channels while controlling for foreign-currency debt burdens. Han and Pyun (2021) deflate firms' total direct exports in the Survey of Business Activities and analyze export and firm-performance responses to industry-specific real effective exchange rates by firm size. They use import values to construct an import share relative to the cost of goods sold as a control. Kwon et al. (2025) use total exports and imports and export and import participation from the same survey to examine heterogeneous responses through interactions between the won–dollar exchange rate and firm size. They also report results using industry-specific effective exchange rates. Because these studies use firm–year trade aggregates, they cannot directly distinguish adjustments in transactions with particular partner countries.

Kim et al. (2026) combine firm financial records with firm–product–destination customs data and show that export-quantity and price responses to foreign-currency debt exposure after the sharp depreciation of the won during the 1997 Asian financial crisis differ by firm size. Using panel data for 2001–2020, they also identify heterogeneity through financial channels using interactions among won–dollar exchange rate changes, foreign-currency debt shares, and firm size. Whereas their study focuses on firms' financial exposure to won–dollar exchange rate shocks, we examine export and import responses to bilateral real exchange rate movements between Korea and its trading partner countries.

In general, when a single country's firm-level data are combined with changes in its currency's exchange rate against the dollar, the direct exchange rate coefficient is absorbed by year fixed effects because the movement is common to all firms. Although our analysis is also restricted to Korean firms, it exploits differences in bilateral real exchange rate movements across trading partner countries. Such movements vary across countries within the same year, allowing us to analyze transactions with specific partners while controlling for common time shocks. Using TEC firm–country–product transactions, we estimate these responses separately for exports and imports and use firm-specific exposure to extend the transaction-level analysis to overall outcomes such as employment and sales.

3 Data and Units of Analysis

3.1 Trade by Enterprise Characteristics and Data Linkage

We use Trade by Enterprise Characteristics (TEC), jointly compiled by Statistics Korea and the Korea Customs Service. The available data span 2010–2024, and the main sample covers 2015–2024, when firm-size information is available. TEC reports export and import values in U.S. dollars and won by trading partner and HS10 product. We use won-denominated values as the dependent variables. For the transaction-level regressions, we aggregate HS10 values to HS6 to link them to the Korean Standard Industrial Classification (KSIC). Product classifications beyond six digits, such as HS10, are defined nationally, whereas HS6 is internationally harmonized. Using HS6 also facilitates linkage to international trade data from UN Comtrade and WITS and comparisons across countries.

The TEC firm identifier is the unique establishment identifier of the firm's representative establishment, which permits linkage to other firm and establishment data from Statistics Korea. Table 1 summarizes linkage for 2023. The first column reports the numbers of observations and establishments in TEC. In the 2023 TEC data provided for this study, 36,293 establishments have observed export activity and 54,341 have observed import activity.

The second column links TEC to the 73,050 establishments in the Mining and Manufacturing Survey (MMS). Whereas TEC covers firms engaged in exports or imports, MMS covers mining and manufacturing establishments with at least ten workers. The 386,480 export observations in the second column amount to approximately 69% of the 561,668 TEC export observations in the first column. The third column reports trade activity for establishments observed in both TEC and MMS. Among MMS establishments, 17,328 have export activity and 21,112 have import activity recorded in TEC.

The final column links the Survey of Business Activities (SBA), MMS, and TEC. In principle, SBA covers incorporated firms with at least 50 regular employees and paid-in capital of at least KRW 300 million, and provides firm identifiers linking establishments to firms. The three-way linkage yields 9,945 establishments belonging to 6,556 firms.

TEC is the primary dataset for our analysis. The firm-level analysis uses establishments linked to MMS. SBA information is not used in the empirical analysis; the linkage statistics are reported to facilitate future research.

An advantage of firm–country–product data is that transactions can be distinguished by partner and product even within the same firm. If only total firm exports are observed, a decline in exports to one country that is offset by a shift toward another country is difficult to identify. TEC allows us to examine responses sequentially at several levels, from firm–country–product transactions to the firm

Table 1. Linkage of TEC and Firm Data in 2023

	TEC	MMS+TEC	TEC+MMS	MMS+SBA+TEC
Panel A. Exports				
N obs	561,668	386,480	330,758	199,793
N est	36,293	73,050	17,328	9,945
N Firms				6,556
Panel B. Imports				
N obs	677,691	395,340	343,402	202,570
N est	54,341	73,050	21,112	9,945
N Firms				6,556

Source: Descriptive statistics for the 2023 linkage in the consolidated results tables dated October 21, 2026. The original table is presented as separate export and import panels. Column headings, statistic labels, and values follow the source; blank cells indicate that the relevant statistic was not tabulated. N obs denotes observations, N est establishments, and N Firms firms.

as a whole. This structure is particularly useful for distinguishing trade expansion from changes in trading structure.

Unlike some other firm–country–product datasets, the available TEC data do not provide transaction quantities. This prevents us from distinguishing price and quantity adjustment in response to exchange rate movements or other changes in external conditions. TEC also does not report firms' domestic locations, limiting spatial analysis based on their geographical distribution within Korea.

3.2 Key Variables and Aggregation Levels

Let f denote a firm, p a product, c a trading-partner country, and t a year. Country c is the destination in the export analysis and the country of origin in the import analysis. X_{fpct} and M_{fpct} are export and import values in won, respectively. Firm–country trade values are obtained by summing over products, and total firm trade is obtained by summing again over countries. $Sales_{ft}$ denotes total sales, including both domestic sales and exports.

The explanatory variables are the bilateral real exchange rate between Korea and trading partner country c, RER_{ct}, and that country's real GDP, RGDP_{ct}. An increase in RER_{ct} denotes a real depreciation of the won against the currency of country c. We use log differences of both variables. A positive exchange rate coefficient therefore indicates that real depreciation of the won is associated with an increase in the dependent variable. Exchange rate variation occurs at the country–year level, rather than at the firm or product level. The fact that multiple firms and products share the same country–year exchange rate is relevant to standard errors and identification. In the empirical analysis below, "exchange rate" refers to this bilateral real exchange rate unless otherwise stated.

The product count is the number of HS6 products a firm trades with a given country. Product concentration is the sum of squared product shares in the firm's trade value with that country. Country concentration is the sum of squared country shares in the firm's total trade value. The former captures product composition within a country, while the latter captures partner-country composition for the firm as a whole. Product and country counts enter in log differences, while both Herfindahl–Hirschman indices (HHIs) enter in level differences. HHI coefficients therefore represent changes in the level of the index, not percentage changes as for log-differenced variables.

Table 2. Key Variables and Units of Analysis

Variable	Observation unit	Definition and transformation
Trade value	Firm–country–product–year	Log difference of export or import value in won
Bilateral real exchange rate	Country–year	An increase denotes real depreciation of the won; log difference
Real GDP	Country–year	Log difference of the trading partner's real GDP
Product count	Firm–country–year	Log difference of the number of HS6 products with positive trade
Product HHI	Firm–country–year	Level difference of concentration based on product shares in trade value
Trade indicator	Firm–country–year	One if trade occurs and zero otherwise; samples split by prior-year status
Employment and wage controls	Firm–year	Firm–year log levels in entry and continuation regressions
Total sales and employment	Firm–year	Log differences in the firm-performance regressions
Country count	Firm–year	Log difference of the number of countries with positive trade
Country HHI	Firm–year	Level difference of concentration based on country shares in trade value
Exchange rate and GDP exposure	Firm–year	Log changes weighted by lagged country trade shares, multiplied by the lagged export- or import-to-sales ratio

4 Empirical Methodology

The empirical analysis proceeds at three levels: firm–country–product, firm–country, and firm. At each level, we examine export and import responses and discuss the scope of identification and statistical inference.

4.1 Firm–Country–Product Analysis

4.1.1 Trade Values

The baseline model directly follows the framework in Li et al. (2015, Section 4.1, equation (2)). We replace their price or quantity outcome with the won-denominated trade value observed in TEC and estimate

$$\Delta \ln V_{fpct} = \mu + \alpha \Delta \ln \text{RER}_{ct} + \beta \Delta \ln \text{RGDP}_{ct} + \xi_{fpc} + \tau_t + \varepsilon_{fpct}. \tag{1}$$

Here, V_{fpct} is X_{fpct} for exports and M_{fpct} for imports. ξ_{fpc} denotes firm–country–product fixed effects, and τ_t denotes year fixed effects. Real GDP growth controls for changes in the trading partner's economic size. For exports, it serves as a proxy for demand conditions; for imports, it controls for economic conditions in the origin country. We do not assume that the GDP coefficients capture the same economic channel in the two trade directions.[1]

The coefficient of interest, α, captures the response of won-denominated trade values to a real depreciation of the won against the currency of the relevant trading partner country. In this log-difference regression, a 1% increase in the bilateral real exchange rate is associated, other things equal, with an approximately α% change in trade value. This coefficient is neither a quantity elasticity nor a foreign-currency price pass-through rate. In particular, won-denominated import expenditure can rise even when quantities fall if won-denominated prices increase at the same time.

Equation (1) is a growth-rate model with fixed effects. Taking differences does not make firm–country–product fixed effects redundant: they allow average growth rates to differ across trading relationships. Year fixed effects absorb annual changes common to all transactions. The coefficient on the bilateral real exchange rate is identified from the remaining variation across trading partners. Standard errors are clustered at the firm level, as in the latest consolidated results tables.

To examine whether export-value sensitivity to exchange rates changes over time, we augment equation (1) with interactions between year indicators and changes in bilateral real exchange rates. Using 2019 as the reference year, we estimate

$$\begin{aligned} \Delta \ln X_{fpct} = \mu + \alpha \Delta \ln \text{RER}_{ct} + \sum_{s \neq 2019} \gamma_s D_{st} \Delta \ln \text{RER}_{ct} \\ + \beta \Delta \ln \text{RGDP}_{ct} + \xi_{fpc} + \tau_t + \varepsilon_{fpct}. \end{aligned} \tag{2}$$

[1] Although we follow the firm–product–country framework of Li et al. (2015), the available TEC data do not contain transaction quantities or unit prices. We therefore focus on export and import values and cannot decompose exchange-rate-related changes in these values into quantity and price adjustment.

D_{st} equals one when $t = s$. The coefficient α is the exchange rate coefficient in the reference year, 2019, and γ_s is the difference in sensitivity between year s and 2019. The total coefficient in year s is thus $\alpha + \gamma_s$, and the pattern of γ_s across years indicates changes in exchange rate sensitivity. As in the baseline model, we control for firm–country–product and year fixed effects and trading-partner real GDP growth. Because year fixed effects absorb common annual shocks, the annual export responses are estimated using differences in bilateral real exchange rate movements across trading partners within each year. Standard errors are clustered at the firm level.

4.1.2 Heterogeneity by Firm Size and Region

For the analysis by firm size, we estimate equation (1) separately for large firms, mid-sized firms, and SMEs. For the regional analysis, we add interactions between changes in bilateral real exchange rates and region indicators, using East Asia as the reference region:

$$\begin{aligned}\Delta \ln X_{fpct} = \mu + \alpha \Delta \ln \text{RER}_{ct} + \sum_{r \neq EA} \gamma_r D_{cr} \Delta \ln \text{RER}_{ct} \\ + \beta \Delta \ln \text{RGDP}_{ct} + \xi_{fpc} + \tau_t + \varepsilon_{fpct}.\end{aligned} \tag{3}$$

D_{cr} equals one if country c belongs to region r. The exchange rate coefficient for East Asia is α, and the total coefficient for another region is $\alpha + \gamma_r$. The confidence interval for γ_r describes uncertainty about the difference from East Asia, not uncertainty about that region's total coefficient. To consider firm size and region jointly, we estimate this model separately for each size group.

Differences in point estimates across firm sizes in the overall sample do not establish statistically significant differences between the coefficients. Moreover, comparing differences relative to East Asia across size groups, as in Figure 4, is not equivalent to comparing regional total effects, because the East Asia main effect itself differs by size. We report the magnitudes of the point estimates and use tests of whether the directly estimated regional interactions differ from zero to describe regional differences within each size group.

4.2 Firm–Country Analysis

4.2.1 Product Composition

At the firm–country level, we analyze adjustment in product counts and product concentration:

$$\Delta y_{fct} = \mu + \alpha \Delta \ln \text{RER}_{ct} + \beta \Delta \ln \text{RGDP}_{ct} + \xi_{fc} + \tau_t + \varepsilon_{fct}, \tag{4}$$

where y_{fct} is either the log product count or the product HHI. Both the export and import specifications include firm–country fixed effects, ξ_{fc}, and year fixed effects, τ_t. An increase in the product count indicates a broader range of traded products, while a decline in the HHI indicates a more dispersed distribution of trade value. These are distinct concepts: concentration can change through shifts in the shares of existing products even if the product count remains unchanged.

4.2.2 Market Entry and Continuation

Our definitions of market entry and continuation and the associated sample split follow Li et al. (2015, Section 5). Let $x_{fct} = 1$ if firm f trades with country c in year t and zero otherwise, with separate indicators for exports and imports. As shown in Table 3, the entry and continuation samples are defined by the previous year's trading status. The entry sample consists of firm–country pairs that did not trade in the previous year ($x_{fc,t-1} = 0$). Current trade ($x_{fct} = 1$) denotes the start or resumption of trade, while no current trade ($x_{fct} = 0$) denotes continued inactivity. The continuation sample consists of pairs that traded in the previous year ($x_{fc,t-1} = 1$). Current trade denotes continuation of the relationship; no current trade denotes termination of trade with that country, or exit from that market. Both analyses therefore use current trading status as the dependent variable, but their samples and the interpretation of the estimated probability differ with lagged trading status.

Table 3. Conditional Samples for Market Entry and Continuation

Outcome	Prior-year condition	Current $x_{fct} = 1$	Current $x_{fct} = 0$
Entry	$x_{fc,t-1} = 0$	Start or resumption of trade	Continued inactivity
Continuation	$x_{fc,t-1} = 1$	Continuation of existing trade	Exit from the market

Trading status is defined at the firm–country–year level. Termination of trade with a particular country is classified as exit from that market even if the firm continues trading with other countries.

Entry under this definition is not the same as a firm's first-ever export transaction or its establishment. It also includes reentry into a country with which the firm previously traded. Continuation likewise refers to the persistence of a trading relationship with a particular country, rather than to firm survival. Within the same conditional sample, the exit probability is the complement of the continuation probability, so a negative continuation response corresponds to a positive exit response.

Following Li et al. (2015), we consider firm–country pairs that traded at least once during the sample period and for which the required explanatory variables are available throughout the period. This does not mean that only continuously trading pairs are included. We construct annual trading indicators for pairs with some trading history and record $x_{fct} = 0$ in years without trade, allowing us to observe entry and termination. Failed data linkages and missing explanatory variables are not treated as zero

trade. Only observations with an identifiable prior-year trading status enter the entry and continuation analyses.

Constructing annual observations for every combination of firms and countries would produce a dataset whose size is proportional to the product of the numbers of firms, countries, and years. Restricting attention to pairs with some trading history avoids generating never-trading combinations and reduces the computational burden of data construction and estimation. The estimated entry probability is consequently conditional on relationships with some observed trading history and cannot be generalized to the probability that any firm enters any foreign market. Li et al. (2015) also acknowledge that this sample restriction may affect the magnitude of estimated effects.

Let $d \in \{0, 1\}$ denote the previous year's trading status. Our specification is

$$\begin{aligned}\Pr(x_{fct} = 1 \mid x_{fc,t-1} = d) = G\big(a_d + \alpha_d \Delta \ln \mathrm{RER}_{ct} + \beta_d \Delta \ln \mathrm{RGDP}_{ct} \\ + \theta_d \ln Emp_{ft} + \phi_d \ln Wage_{ft} + \eta_c + \tau_t\big).\end{aligned} \tag{5}$$

G is the standard normal cumulative distribution function in the probit model and the logistic function in the logit model. The linear probability model (LPM) uses a linear specification with the same explanatory variables. Entry corresponds to $d = 0$ and continuation to $d = 1$. Employment and wages are measured at the firm–year level and thus carry the subscript ft. We include country fixed effects, η_c, and year fixed effects, τ_t. Following the consolidated results tables, standard errors are clustered by destination for exports and by origin country for imports.

4.3 Firm-Level Analysis

To link bilateral real exchange rate movements to overall firm outcomes, we construct exposure measures based on lagged trading structures. Let $q \in \{X, M\}$ denote the trade direction and V^q the corresponding trade value. Country weights and exposure to macroeconomic variables are defined as

$$w^q_{fc,t-1} = \frac{V^q_{fc,t-1}}{V^q_{f,t-1}}, \tag{6}$$

$$E^{k,q}_{ft} = \sum_c w^q_{fc,t-1} \Delta \ln k_{ct}, \qquad k \in \{\mathrm{RER}, \mathrm{RGDP}\}. \tag{7}$$

To account for the importance of exports or imports relative to total firm sales, we define

$$T^{k,q}_{ft} = E^{k,q}_{ft}\, s^q_{f,t-1}, \qquad s^q_{f,t-1} = \frac{V^q_{f,t-1}}{Sales_{f,t-1}}. \tag{8}$$

For imports, we apply the same construction using lagged origin-country import shares and the lagged ratio of imports to total sales.

The firm-outcome model is

$$\Delta y_{ft} = \beta T_{ft}^{\text{RER},q} + \theta T_{ft}^{\text{RGDP},q} + \delta_f + \tau_t + u_{ft}. \tag{9}$$

Employment, total sales, and the number of trading countries enter in log differences. Both export and import country HHIs enter in level differences. The model includes firm and year fixed effects, with standard errors clustered at the firm level. Because exposure varies across firms with their trading structures, β cannot be interpreted as a simple exchange rate elasticity common to all firms.

5 Export Results

This section examines exporters' responses to exchange rate movements using the models and sequence of analysis presented in Section 4. We first estimate export-value responses at the firm–country–product level and examine variation across years, firm sizes, and destination regions. We then study product composition within destinations and export market entry and continuation at the firm–country level. Next, we relate firm-specific exchange rate exposure to sales, employment, and trading-partner composition. Finally, we bring these results together to discuss exporters' main adjustment channels.

5.1 Firm–Country–Product Analysis

5.1.1 Won-Denominated Export Values and Responses by Firm Size

Table 4 reports estimates of equation (1) for the full sample and by firm size, with won-denominated exports at the firm–destination–product level as the outcome.[2] In the full sample, the bilateral real exchange rate coefficient is 0.229, with a standard error of 0.022. Other things equal, a 1% real depreciation of the won is associated with an approximately 0.229% increase in the won-denominated export value of the transaction. The real GDP coefficient is also positive, indicating that trading-partner economic growth is associated with higher exports.

The positive relationship between bilateral real exchange rates and export values remains after controlling for common annual shocks and differences in average growth across trading relationships. Unlike analyses based on total firm exports, this specification separately identifies responses in trans-

[2]The baseline year fixed effects control for shocks common to all transactions but do not fully absorb product-specific annual shocks arising from industry cycles or changes in demand and supply conditions. Appendix A addresses this issue by estimating equation (10) with firm–country and HS6–year fixed effects. In Table 12, the exchange rate coefficient for exports is 0.185. Although smaller than in the baseline, it remains positive.

Table 4. Bilateral Real Exchange Rates and Won-Denominated Exports: Overall and by Firm Size

	All firms	Large firms	Mid-sized firms	SMEs
$\Delta \ln \text{RER}_{ct}$	0.229^{***}	0.207^{***}	0.224^{***}	0.238^{***}
	(0.022)	(0.076)	(0.056)	(0.023)
$\Delta \ln \text{RGDP}_{ct}$	1.417^{***}	1.814^{***}	1.449^{***}	1.264^{***}
	(0.070)	(0.262)	(0.139)	(0.075)
Observations	2,200,635	331,467	440,915	1,428,110
R^2	0.339	0.354	0.357	0.352
Firm–country–product FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

The dependent variable is the log difference in won-denominated exports at the firm–destination–HS6 level. Standard errors are clustered at the firm level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

actions with particular trading partners. Changes in won-denominated export values, however, incorporate foreign-currency price changes, quantity adjustment, and currency conversion effects. The coefficient of 0.229 cannot therefore be interpreted as an export-quantity elasticity. Nor does the fact that export-value growth is smaller than the rate of real depreciation reveal the separate price and quantity responses.

The magnitude of the export-value response is similar to that reported for Chinese firms by Li et al. (2015). In their baseline estimates, a 1% real depreciation of the renminbi raises renminbi-denominated export unit values by 0.035% and quantities by 0.226%. Adding these coefficients implies an export-value elasticity of approximately 0.261, close to our estimate of 0.229. Their results indicate that quantity adjustment contributes more than unit-value adjustment to Chinese firms' export-value response. Because we cannot distinguish prices from quantities, however, similar value responses need not imply identical adjustment mechanisms.

The exchange rate coefficients are 0.207 for large firms, 0.224 for mid-sized firms, and 0.238 for SMEs. All three are positive, and the point estimate is largest for SMEs. This ordering is consistent with the possibility of larger responses among smaller firms. Testing whether each coefficient differs from zero is not, however, equivalent to testing whether the coefficients differ across size groups. We therefore do not claim that the cross-size differences are statistically established. Section 5.1.2 presents regional interactions estimated separately by firm size and shows that the pattern of larger SME point estimates also appears in the regional results.

Whether exporters' exchange rate sensitivity changes over time matters for the intensity and form of policy responses. If firms react differently to exchange rate movements of the same magnitude, policies based on historical average responses may no longer suit current conditions. We therefore examine annual changes in sensitivity. Figure 2 plots the year interaction coefficients from equation (2)

and their 95% confidence intervals. The exchange rate coefficient in the reference year, 2019, is 0.255. Each point shows how sensitivity in that year differs from the 2019 coefficient. All reported confidence intervals include zero, providing no statistically significant evidence of a difference from 2019. Nor do the estimates show a consistently increasing or decreasing pattern over time. These results therefore do not establish a change in the rationale for adjusting policy intensity or design in response to changing exchange rate sensitivity.

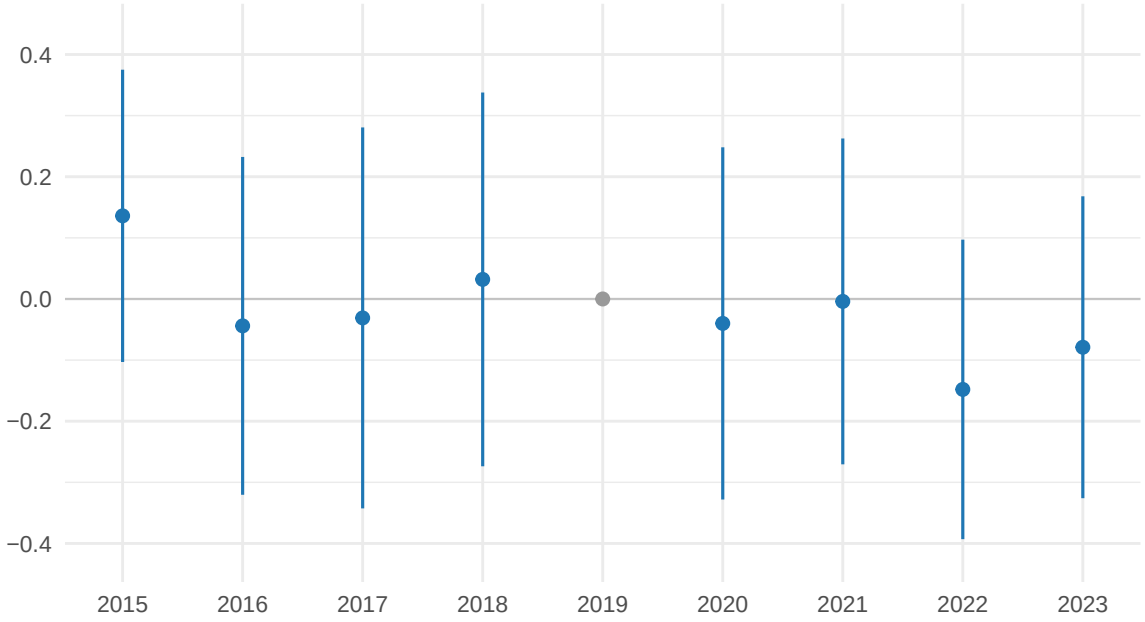


Figure 2. Export Responses to Bilateral Real Exchange Rates: Differences from 2019

Points show year interaction coefficients γ_s; lines show $\gamma_s \pm 1.96SE$. The reference year, 2019, is plotted at zero without a confidence interval. Controls include firm–country–product and year fixed effects and trading-partner real GDP growth. Standard errors are clustered at the firm level. Only years reported in Appendix Table 14 are shown.

5.1.2 Regional Responses and the Role of Firm Size

Figure 3 presents the regional interaction coefficients from equation (3), with East Asia as the reference region, and their 95% confidence intervals. The East Asia main effect is 0.393, with a standard error of 0.059. The Europe interaction is -0.333, indicating a smaller response than in East Asia. Adding the two coefficients gives a point estimate of $0.393 - 0.333 = 0.060$ for Europe's total effect. The negative interaction should therefore not be interpreted as evidence that depreciation of the won reduces exports to Europe.

Regional differences reveal heterogeneity obscured by a single average coefficient on the bilateral real exchange rate. The same change in the won's value may generate different trade responses depending on product composition, competitive conditions, and pricing practices in each market. Interactions between exchange rates and region indicators alone cannot, however, identify a particular mechanism.

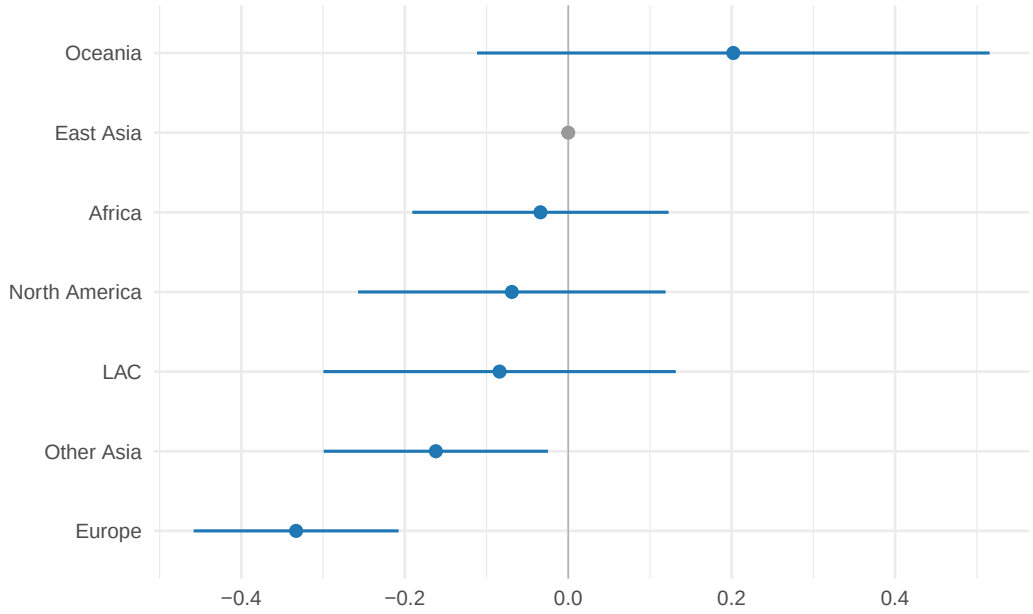


Figure 3. Export Responses to Bilateral Real Exchange Rates: Differences from East Asia

Points show the interaction coefficients γ_r in equation (3); lines show $\gamma_r \pm 1.96SE$. East Asia is the reference group and is plotted at zero. Its main effect is 0.393 (standard error 0.059). LAC denotes Latin America and the Caribbean. The figure does not show confidence intervals for regional total effects. Standard errors are clustered at the firm level.

Assessing the significance of Europe's total effect also requires a standard error that incorporates the covariance between the main effect and the interaction.

Figure 4 reports regional interactions estimated by applying equation (3) separately to large firms, mid-sized firms, and SMEs. Each panel uses East Asia within the relevant size group as its reference. The East Asia main effects are -0.024 for large firms, 0.294 for mid-sized firms, and 0.473 for SMEs. As in the overall results by firm size, the SME point estimate is larger, and the gap is more pronounced here.

Within SMEs, differences between East Asia and other regions are more evident. The 95% confidence intervals for the Africa, Europe, North America, and Other Asia interactions all exclude zero. Responses in these regions are therefore significantly smaller than the SME response in East Asia at the 5% level. The intervals for Latin America and the Caribbean and for Oceania include zero. The numerical intervals appear in Appendix Table 13.

5.2 Firm–Country Analysis

5.2.1 Product Composition within Destinations

Table 5 reports estimates of equation (4) for the number and concentration of export products within the same destination. The exchange rate coefficient for the log difference in product count is 0.029

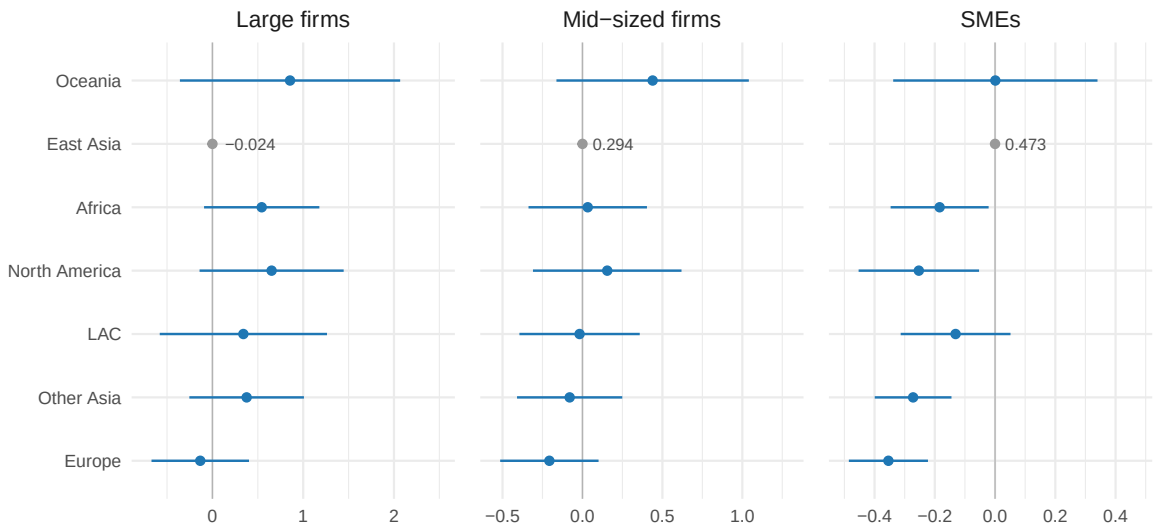


Figure 4. Export Responses to Bilateral Real Exchange Rates: Regional Differences by Firm Size

Each panel shows differences relative to East Asia from a separate regression for that size group. Lines denote 95% confidence intervals for interaction coefficients. Numbers beside the East Asia points are the corresponding main effects. Horizontal-axis ranges differ across panels, so visual positions alone should not be used to compare size groups. LAC denotes Latin America and the Caribbean. The reference points at zero do not display confidence intervals for the main effects. Standard errors are clustered at the firm level.

and significant at the 1% level. A 1% real depreciation of the won is associated with an approximately 0.029% increase in the number of products exported by the same firm to the same destination. The coefficient for the change in product HHI is -0.007, significant at the 5% level. Depreciation of the won is thus associated with more products and a less concentrated distribution of trade value.

Table 5. Bilateral Real Exchange Rates and Export Product Composition

	Log change in product count	Change in product HHI
$\Delta \ln \text{RER}_{ct}$	0.029^{***}	-0.007^{**}
	(0.009)	(0.003)
$\Delta \ln \text{RGDP}_{ct}$	0.277^{***}	-0.040^{***}
	(0.026)	(0.010)
Observations	981,956	981,956
R^2	0.166	0.161
Firm–destination FE	Yes	Yes
Year FE	Yes	Yes
Clustering	Firm	Firm

The unit of analysis is the firm–destination–year. HHI enters in level differences, so its coefficient is not interpreted as a percentage change. Standard errors are clustered at the firm level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

These results suggest that exchange rate responses need not be confined to expanding the value of existing products. Firms may also add products in destinations they already serve or adjust sales shares across existing products.

5.2.2 Export Market Entry and Continuation

Table 6 reports probit, logit, and LPM estimates of the entry and continuation models in equation (5), splitting firm–destination observations according to whether trade occurred in the previous year. The probit exchange rate coefficients are 0.006 for entry and -0.014 for continuation; the corresponding logit coefficients are 0.023 and -0.028.[3] The LPM coefficients, 0.003 for entry and -0.004 for continuation, have the same signs. None of the models provides statistically significant evidence of an exchange rate effect.

Table 6. Export Market Entry and Continuation

	Probit		Logit		LPM	
	Entry	Continuation	Entry	Continuation	Entry	Continuation
$\Delta \ln \text{RER}_{ct}$	0.006	-0.014	0.023	-0.028	0.003	-0.004
	(0.104)	(0.061)	(0.186)	(0.102)	(0.020)	(0.021)
$\Delta \ln \text{RGDP}_{ct}$	0.503***	0.691***	0.955***	1.160***	0.102***	0.243***
	(0.154)	(0.201)	(0.283)	(0.341)	(0.032)	(0.069)
$\ln Emp_{ft}$	0.034***	0.167***	0.063***	0.289***	0.008***	0.050***
	(0.002)	(0.008)	(0.004)	(0.015)	(0.0005)	(0.002)
$\ln Wage_{ft}$	0.124***	0.174***	0.229***	0.292***	0.027***	0.056***
	(0.008)	(0.030)	(0.014)	(0.050)	(0.002)	(0.009)
Observations	1,635,660	887,789	1,635,660	887,789	1,635,660	887,789
Destination FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Destination	Destination	Destination	Destination	Destination	Destination

Standard errors are clustered by destination. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

These findings are consistent with the relationship between depreciation of the won and higher export values being more apparent in adjustments within existing relationships than in entry into or continuation in a particular market. The coefficients on trading-partner GDP, firm employment, and wages are positive, indicating conditional relationships between market participation, expanding destination economies, and firm characteristics.

5.3 Firm-Level Analysis

Table 7 reports estimates of equation (9) using export-side exchange rate exposure constructed from lagged destination shares and the export-to-total-sales ratio, as defined in equations (6)–(8). The exchange rate exposure coefficient for the log difference in total sales is 0.054, with a standard error of 0.031, and is significant at the 10% level. Firms more exposed to depreciation of the won exhibit

[3] Probit and logit coefficients are not marginal effects and cannot be read directly as changes in probabilities.

higher total sales growth, although the evidence is limited. Total sales include both domestic sales and exports, so this result does not describe domestic sales alone.

Table 7. Export-Side Exchange Rate Exposure and Firm Outcomes

	Employment	Total sales	Country count	Country HHI
Exchange rate exposure	−0.022	0.054*	−0.034	0.000
	(0.023)	(0.031)	(0.025)	(0.004)
GDP exposure	−0.011	0.009	0.031***	−0.016*
	(0.025)	(0.060)	(0.010)	(0.009)
Observations	116,561	116,674	116,909	116,098
R^2	0.213	0.222	0.148	0.138

The model includes only export-side exposure. Exposure measures follow equation (8); exchange rate exposure is constructed from changes in bilateral real exchange rates. Employment, total sales, and country count enter in log differences; HHI enters in level differences. Firm and year fixed effects are included. Standard errors are clustered at the firm level. The value 0.000 is rounded. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The coefficients for employment and the number of trading countries are -0.022 and -0.034, respectively, and are not significant. The coefficient for the change in country HHI is 0.000 when rounded to three decimal places and is also insignificant. The results therefore provide no evidence that higher export values translate directly into firm-wide employment expansion or destination diversification. In particular, lower product concentration within a destination and changes in country concentration for the firm as a whole are different adjustments and need not coincide.

The GDP exposure coefficients for employment and total sales are -0.011 and 0.009, respectively, and are not significant. In contrast, the coefficient for the number of trading countries is 0.031 and significant at the 1% level, while the coefficient for the change in country HHI is -0.016 and significant at the 10% level. Firms more exposed to economic growth in their lagged export destinations tend to increase the number of trading countries and reduce destination concentration. Changes in overall destination composition are thus observed in conjunction with destination economic growth rather than exchange rate exposure. Because GDP exposure is a proxy for demand conditions, these coefficients should not be interpreted as causal effects of a specific demand channel.

Interpreting exposure coefficients requires attention to firms' export shares. For the same bilateral real exchange rate movement, a firm with a higher export-to-total-sales ratio experiences a larger change in the exposure measure in equation (8). The coefficient of 0.054 therefore does not mean that a 1% real depreciation of the won raises sales by 0.054% for every firm. Total sales growth may also be partly mechanically related to higher won-denominated export values and should be distinguished from an expansion of domestic production or employment.

5.4 Export Adjustment Channels

Taken together, the export results indicate that responses to exchange rate movements are more apparent in values and product composition within existing relationships than in trading-partner composition or employment. Table 4 associates a 1% real depreciation of the won with an approximately 0.229% increase in won-denominated exports at the firm–destination–product level. Combined with the higher product counts and lower product HHI in Table 5, this suggests that firms adjust not only the value of existing products but also the range of products and their sales shares within established destinations. In particular, lower product concentration within a destination shows that export diversification can occur without entry into new countries.

By contrast, the baseline models in Table 6 yield no significant exchange rate responses for export market entry or continuation. Table 7 likewise provides limited evidence of changes in country counts or country concentration associated with exchange rate exposure. The firm-wide employment response is not significant, and the total-sales exposure coefficient of 0.054 is significant only at the 10% level. It is therefore difficult to conclude that exchange-rate-related adjustments in export values and product composition lead quickly to broader market coverage or greater employment. Distinguishing product diversification within a country from destination diversification across the firm helps identify the level at which adjustment occurs.

This pattern is consistent with fixed and adjustment costs associated with market development and employment expansion. Entry into a new destination may require information about local demand and regulations, a search for trading partners, and the establishment of distribution networks and contractual relationships. Hiring and training are also costly, so firms may consider the persistence of future demand and profits rather than expand markets or employment in response to a temporary improvement in profitability. If exchange rate movements are perceived as temporary, firms may postpone these costly decisions and instead adjust export values and product composition through existing partners and distribution networks. In that case, adjustment takes place within established relationships rather than through expansion into new markets or additional production capacity.

6 Import Results

Imports are a major source of raw materials and intermediate inputs for production. Changes in import costs and sourcing structures induced by exchange rate movements may therefore affect firms' production and employment. Understanding firms' responses to exchange rate movements requires examining adjustment on the import-sourcing side as well as the export-sales side. This section applies

the models and sequence of analysis in Section 4 to importers. We extend the export specifications of Li et al. (2015) to import transactions; their original study does not analyze imports. We first estimate import-value responses at the firm–country–product level, then examine product composition within origin countries and import market entry and continuation at the firm–country level. We next relate firm-specific exchange rate exposure to sales, employment, and partner-country composition. Finally, we discuss importers' main adjustment channels and compare responses in sales and sourcing markets.

6.1 Firm–Country–Product Analysis

6.1.1 Won-Denominated Import Values

The first column of Table 8 applies equation (1) to won-denominated imports at the firm–origin–product level.[4] The bilateral real exchange rate coefficient is 0.512, with a standard error of 0.035, and is significant at the 1% level. Other things equal, a 1% real depreciation of the won is associated with an approximately 0.512% increase in the transaction's won-denominated import value. The real GDP coefficient is also positive, indicating that economic growth in the origin country is associated with higher import values.

Table 8. Bilateral Real Exchange Rates, Won-Denominated Imports, and Product Composition

	Log change in import value	Log change in product count	Change in product HHI
$\Delta \ln \text{RER}_{ct}$	0.512^{***}	-0.017	0.016^{***}
	(0.035)	(0.015)	(0.006)
$\Delta \ln \text{RGDP}_{ct}$	0.340^{***}	0.396^{***}	-0.065^{***}
	(0.093)	(0.045)	(0.018)
Observations	2,166,551	792,827	792,827
R^2	0.361	0.180	0.175
Unit of analysis	Firm–origin–HS6	Firm–origin	Firm–origin
Relationship FE	Firm–origin–HS6	Firm–origin	Firm–origin
Year FE	Yes	Yes	Yes
Clustering	Firm	Firm	Firm

Countries are import origins, import values are denominated in won, and products are defined at HS6. The first column estimates equation (1); the remaining columns estimate equation (4). Standard errors are clustered at the firm level.

An increase in won-denominated import values must be distinguished from an expansion of import quantities. Depreciation of the won can raise the unit cost in won even if the foreign-currency price remains fixed. If the price increase exceeds the reduction in purchased quantities, won-denominated ex-

[4]The positive relationship also remains when equation (10) is estimated with firm–country and HS6–year fixed effects to account for product-specific annual shocks. The exchange rate coefficient for imports is 0.450. Appendix A and Table 12 describe the alternative specification and its results.

penditure rises. Import values can also increase when firms maintain purchases of essential inputs. The coefficient of 0.512 is therefore a value response combining price, quantity, and currency conversion effects; it cannot be interpreted as an import-quantity elasticity or as a measure of input substitution.

6.2 Firm–Country Analysis

6.2.1 Product Composition within Origin Countries

The second and third columns of Table 8 use equation (4) to estimate responses in import product counts and concentration at the firm–origin level. The exchange rate coefficient for the log difference in product count is -0.017 and is not statistically significant. The coefficient for the change in product HHI is 0.016 and significant at the 1% level. Thus, while there is no clear evidence that depreciation of the won changes the range of imported products, expenditure within the same origin country becomes more concentrated in particular products.

Expenditure shares can change even when product counts do not. For example, if purchases of some products fall relatively more, or the won-denominated prices of particular products rise more sharply, value-based concentration can increase without any change in the number of products. These results reveal changes in expenditure composition within existing sourcing relationships, but they do not establish a decline in input diversification measured in quantities. Product composition within an origin country and origin-country composition for the firm as a whole are distinct adjustments. We examine the latter through market entry and continuation and the firm-level analysis.

6.2.2 Import Market Entry and Continuation

Table 9 reports probit, logit, and LPM estimates of equation (5), splitting firm–origin observations by prior-year trading status. The exchange rate coefficients for entry are positive but insignificant: 0.109 for probit, 0.210 for logit, and 0.025 for the LPM. The continuation coefficients are -0.079, -0.130, and -0.026, respectively, all significant at the 5% level. Across all three models, depreciation of the won is consistently associated with a lower probability of continuing trade with an existing origin country. Equivalently, firms become more likely to discontinue imports from a country with which they traded in the previous year. This termination concerns a particular origin-country relationship and does not imply that the firm ceases importing altogether.

Table 9. Import Market Entry and Continuation

	Probit		Logit		LPM	
	Entry	Continuation	Entry	Continuation	Entry	Continuation
$\Delta \ln \text{RER}_{ct}$	0.109	-0.079^{**}	0.210	-0.130^{**}	0.025	-0.026^{**}
	(0.085)	(0.037)	(0.153)	(0.063)	(0.018)	(0.013)
$\Delta \ln \text{RGDP}_{ct}$	0.176	0.104	0.345	0.162	0.040	0.051^{**}
	(0.136)	(0.067)	(0.254)	(0.111)	(0.030)	(0.023)
$\ln Emp_{ft}$	0.050^{***}	0.165^{***}	0.093^{***}	0.270^{***}	0.012^{***}	0.054^{***}
	(0.003)	(0.012)	(0.006)	(0.021)	(0.0008)	(0.003)
$\ln Wage_{ft}$	0.135^{***}	0.305^{***}	0.251^{***}	0.508^{***}	0.030^{***}	0.105^{***}
	(0.009)	(0.029)	(0.016)	(0.043)	(0.002)	(0.013)
Observations	1,552,154	683,507	1,552,154	683,507	1,552,184	683,539
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Origin	Origin	Origin	Origin	Origin	Origin

The entry sample did not trade in the previous year; the continuation sample did. Probit and logit coefficients are not marginal effects. Standard errors are clustered by origin country.

6.3 Firm-Level Analysis

Table 10 estimates equation (9) using import-side exchange rate exposure based on lagged origin-country shares and the import-to-total-sales ratio, as defined in equations (6)–(8). In the model including only import-side exposure, the exchange rate exposure coefficient for the log difference in total sales is 0.391 and significant at the 1% level. The coefficient for the log difference in employment is -0.017 and significant at the 5% level. Firms with greater import exposure to depreciation of the won thus exhibit higher total sales and lower employment.

Table 10. Import-Side Exchange Rate Exposure and Firm Outcomes

	Employment	Total sales	Country count	Country HHI
Exchange rate exposure	-0.017^{**}	0.391^{***}	-0.015	0.006^{**}
	(0.008)	(0.064)	(0.012)	(0.003)
GDP exposure	0.026^{**}	0.649^{***}	0.023	-0.010^{**}
	(0.012)	(0.136)	(0.019)	(0.005)
Observations	134,834	135,078	134,838	134,417
R^2	0.216	0.232	0.129	0.123

The model includes only import-side exposure. Exposure measures follow equation (8); exchange rate exposure is constructed from changes in bilateral real exchange rates. Employment, total sales, and country count enter in log differences; HHI enters in level differences. Firm and year fixed effects are included. Standard errors are clustered at the firm level.

The exchange rate exposure coefficient for the number of trading countries is -0.015 and insignificant, whereas the coefficient for the change in country HHI is 0.006 and significant at the 5% level. Expenditure can therefore become more concentrated in particular countries without a clear change in

the number of origin countries. Country concentration for the firm as a whole and product concentration within an origin country differ in both their units of analysis and samples, so their coefficients should not be added or compared as magnitudes of the same adjustment.

The GDP exposure coefficients for employment and total sales are 0.026 and 0.649, significant at the 5% and 1% levels, respectively. The coefficient for the change in country HHI is -0.010 and significant at the 5% level. The country-count coefficient is 0.023 and insignificant. Firms more exposed to economic growth in their lagged import origins tend to increase employment and total sales and reduce origin-country concentration. In the import analysis, however, GDP exposure is introduced as a control for economic conditions in origin countries, so its coefficients should not be interpreted as causal effects of a particular channel.

Total sales include both domestic sales and exports, and importing firms may also export. The positive sales response cannot therefore readily be attributed to a single channel arising from higher import costs or interpreted as an expansion of real sales volumes. The negative employment response is consistent with adjustment to cost pressures, but the analysis does not directly identify that mechanism. Exposure also varies with firms' trading structures and import shares, so these coefficients are not simple exchange rate elasticities common to all firms.

Because export and import exposure differ in their distributions and samples, the ratio of the export sales coefficient, 0.054, to the import sales coefficient, 0.391, cannot be used to compare the magnitudes of the two channels.

6.4 Import Adjustment Channels

The import results indicate that real depreciation of the won is associated with higher won-denominated expenditure, greater concentration across sourcing products and countries, and the termination of some existing relationships. Within an origin country, value-based product concentration rises without a significant change in product count. At the firm level, country concentration rises without a significant change in the number of trading countries. Understanding firms' responses therefore requires examining the reallocation of expenditure shares within existing trade, as well as the numbers of sourcing products and countries. These results are consistent with the possibility that higher sourcing costs following depreciation of the won affect firms' selection and continuation of trading relationships. If firms maintain purchases of essential goods despite price increases while reducing purchases of other products or from other origins, higher expenditure in continuing relationships can coexist with greater product and country concentration and the termination of some relationships.

At the firm level, greater import-side exchange rate exposure is associated with lower employment and higher total sales. Because higher won-denominated sales may reflect price changes, however, they cannot readily be interpreted as an expansion of production or sales volumes.

7 Conclusion and Policy Implications

This study combines firm–country–product transactions from TEC with bilateral real exchange rates between Korea and its trading partner countries to analyze Korean firms' export and import responses. By exploiting bilateral real exchange rate movements that differ across trading partner countries within the same year, we estimate transaction responses while controlling for common time shocks and extend the analysis to product composition, market participation, and overall firm outcomes. This approach distinguishes adjustment within trading relationships from changes in the relationships themselves, which are obscured in firms' aggregate exports and imports.

Table 11 summarizes the baseline export and import findings. A 1% real depreciation of the won is associated with increases of approximately 0.229% in export values and 0.512% in import values, both measured in won. Beyond trade values, however, adjustment differs between exports and imports. Exporters expand product scope and reduce concentration within the same destination, with no significant responses in market entry, continuation, or employment. Importers instead increase product concentration within the same origin and become less likely to maintain existing origin-country relationships. Lower employment and greater country concentration also appear at the firm level. One interpretation is that depreciation of the won creates room to lower foreign-currency selling prices for exports, whereas it raises the won-denominated cost of imports even if foreign-currency prices remain unchanged. Thus, increases in both won-denominated exports and imports may reflect distinct mechanisms—changes in export price competitiveness and higher import-sourcing costs—and should not be interpreted as equivalent expansions of real activity.

Exporters' responses are more evident in values and product composition within existing relationships. This is consistent with the possibility that, when firms view exchange rate movements as temporary, market-development and workforce-adjustment costs lead them to defer entry into new markets or employment expansion. For imports, the termination of some relationships accompanies higher expenditure concentration, suggesting that firms may adjust their trading structures in response to sourcing-cost pressures. Considering these different responses in sales and sourcing markets is important for understanding firms' exchange rate exposure.

Policy assessment should consider trading partners, product composition, and import dependence alongside firm size and total trade. The variation in responses across firm sizes and regions suggests that

Table 11. Comparison of Main Export and Import Responses

Outcome	Exports	Imports
Trade value in won	Positive: 0.229	Positive: 0.512
Product count	Positive and significant	Negative point estimate; no significant evidence
Product HHI	Negative and significant	Positive and significant
Market entry	No significant evidence	No significant evidence
Termination of existing trade	No significant evidence	Positive; significant at 5%
Total sales	Positive; significant at 10%	Positive; significant at 1%
Employment	No significant evidence	Negative; significant at 5%

Firm outcomes are estimated in models including only the exchange rate and GDP exposure measures for the relevant trade direction. Entry and continuation models include country and year fixed effects. Termination of existing trade is inferred from a decline in the continuation probability and refers to the end of a relationship with a particular trading partner.

firms cannot adequately be assessed using a single average exchange rate elasticity. For firms that both export and import, changes in export-market returns and sourcing costs can operate simultaneously. Understanding exposure in both directions is therefore important when assessing the need for support.

The design of support can also be considered in relation to the observed level of adjustment. For exports, relevant costs may arise in maintaining existing relationships and expanding product scope. For imports, relevant burdens may stem from dependence on specific products or countries and disruptions to sourcing relationships. This study does not directly estimate the effects of particular support policies. Future research should examine how programs such as export vouchers affect export activity and relationship continuation during periods of depreciation of the won, and whether their effects differ from those in normal periods.

The study has several limitations. First, the available data report trade values only, preventing us from distinguishing price and quantity effects. If quantity data become available, they could be combined with trade values to construct unit values and analyze how exchange rate movements affect firms' pricing and quantity adjustment.

Second, the sample includes a period of broad weakness in the won. The linear models assume symmetric responses to appreciation and depreciation, and we do not separately test for differences between the two. Results described in terms of a decline in the won's value should therefore not be assumed to reverse mechanically during appreciation. Future work should distinguish periods of appreciation and depreciation and examine asymmetries in firms' price, quantity, and trading-relationship adjustment.

References

Amiti, M., Itskhoki, O., and Konings, J. (2014). Importers, exporters, and exchange rate disconnect. *American Economic Review*, 104(7):1942–1978.

Berman, N., Martin, P., and Mayer, T. (2012). How do different exporters react to exchange rate changes? *The Quarterly Journal of Economics*, 127(1):437–492.

Blaum, J. (2024). Global firms in large devaluations. *The Quarterly Journal of Economics*, 139(4):2427–2474.

Chen, N. and Juvenal, L. (2016). Quality, trade, and exchange rate pass-through. *Journal of International Economics*, 100:61–80.

Dai, M. and Xu, J. (2017). Firm-specific exchange rate shocks and employment adjustment: Evidence from China. *Journal of International Economics*, 108:54–66.

De Gregorio, J., García, P., Luttini, E., and Rojas, M. (2024). From dominant to producer currency pricing: Dynamics of Chilean exports. *Journal of International Economics*, 149:103934.

Devereux, M. B., Dong, W., and Tomlin, B. (2017). Importers and exporters in exchange rate pass-through and currency invoicing. *Journal of International Economics*, 105:187–204.

Devereux, M. B., Dong, W., and Tomlin, B. (2025). Trade flows and exchange rates: Importers, exporters and products. *Journal of International Economics*, 154:104044.

Eaton, J., Eslava, M., Jinkins, D., Krizan, C. J., and Tybout, J. (2025). A search and learning model of export dynamics. *Journal of International Economics*, 157:104155.

Gopinath, G., Boz, E., Casas, C., Díez, F. J., Gourinchas, P.-O., and Plagborg-Møller, M. (2020). Dominant currency paradigm. *American Economic Review*, 110(3):677–719.

Han, M. and Pyun, J. H. (2021). Effects of exchange rate changes on Korean export firms by firm size. *Journal of Market Economy*, 50(1):99–127. In Korean.

Kim, J., Lee, A. S., and Lee, S. (2026). Are exporters naturally hedged? corporate dollar debt and global trade. Working paper, this version January 2026.

Kwon, S., Park, D., and Lee, S. H. (2025). Heterogeneous responses of exporting firms to exchange rate changes. *The Korean Journal of Industrial Organization*, 33(4):27–68. In Korean.

Lee, K. Y. and Naknoi, K. (2024). Exchange rates, invoicing currencies and the margins of exports. *Journal of International Money and Finance*, 141:103016.

Li, H., Ma, H., and Xu, Y. (2015). How do exchange rate movements affect Chinese exports? — a firm-level investigation. *Journal of International Economics*, 97(1):148–161.

Son, M. (2023). Dominant currency pricing: Evidence from Korean exports. BOK Working Paper 2023-3, Bank of Korea.

A Trade Values with Alternative Fixed Effects

The firm–country–product fixed effects in equation (1) control for differences in average growth across trading relationships, while year fixed effects control for annual shocks common to all transactions. They do not, however, fully capture changes over time that affect only particular products or differ across products. Such shocks are relevant to Korea's trading structure, in which industry cycles and demand and supply conditions may evolve differently. We therefore include HS6–year fixed effects to examine whether the relationship between bilateral real exchange rates and trade values remains after controlling for annual changes common to each product.

The specification with firm–country and HS6–year fixed effects is

$$\Delta \ln V_{fpct} = \mu + \alpha \Delta \ln \text{RER}_{ct} + \beta \Delta \ln \text{RGDP}_{ct} + \xi_{fc} + \tau_{pt} + \varepsilon_{fpct}. \tag{10}$$

Here, p denotes an HS6 product, and V_{fpct} is X_{fpct} for exports and M_{fpct} for imports. τ_{pt} controls for common annual shocks within products. This specification replaces the baseline firm–country–product fixed effects, ξ_{fpc}, with firm–country fixed effects, ξ_{fc}, and replaces year fixed effects with HS6–year fixed effects. It therefore differs from a specification that simply adds controls while retaining the baseline fixed effects.

In Table 12, the bilateral real exchange rate coefficients are 0.185 for exports and 0.450 for imports, both significant at the 1% level. Although smaller than the baseline estimates of 0.229 and 0.512, they preserve the positive relationship between real depreciation of the won and won-denominated trade values.

Table 12. Export and Import Value Responses with Alternative Fixed Effects

	Exports	Imports
$\Delta \ln \text{RER}_{ct}$	0.185^{***}	0.450^{***}
	(0.022)	(0.031)
$\Delta \ln \text{RGDP}_{ct}$	1.333^{***}	0.239^{***}
	(0.066)	(0.090)
Observations	2,200,635	2,166,551
R^2	0.136	0.102
Firm–country FE	Yes	Yes
HS6–year FE	Yes	Yes

The dependent variable is the log difference in won-denominated exports or imports at the firm–country–HS6 level. Estimates are from equation (10), with standard errors clustered at the firm level.

B Confidence Intervals for Regional Interactions by Firm Size

Table 13. 95% Confidence Intervals for Regional Interactions Relative to East Asia

Region	Large firms	Mid-sized firms	SMEs
Africa	[−0.092, 1.178]	[−0.337, 0.403]	[−0.347, −0.021]
Europe	[−0.671, 0.403]	[−0.515, 0.101]	[−0.485, −0.223]
Latin America and the Caribbean	[−0.580, 1.262]	[−0.394, 0.358]	[−0.313, 0.051]
North America	[−0.142, 1.446]	[−0.310, 0.620]	[−0.453, −0.053]
Oceania	[−0.358, 2.068]	[−0.163, 1.041]	[−0.338, 0.340]
Other Asia	[−0.254, 1.008]	[−0.409, 0.249]	[−0.399, −0.145]

C Annual Export Responses

Table 14 reports estimates of equation (2), including the annual interaction coefficients and standard errors plotted in Figure 2 in Section 5.1.1. The specification includes firm–country–product and year fixed effects. The exchange rate coefficient for the reference year, 2019, is 0.255. None of the reported annual interaction coefficients is statistically significant.

Table 14. Annual Export Responses to Bilateral Real Exchange Rates

	Estimate
$\Delta \ln \text{RER}_{ct}$	0.255^{**}
	(0.114)
2015 interaction	0.136
	(0.122)
2016 interaction	-0.044
	(0.141)
2017 interaction	-0.031
	(0.159)
2018 interaction	0.032
	(0.156)
2020 interaction	-0.040
	(0.147)
2021 interaction	-0.004
	(0.136)
2022 interaction	-0.148
	(0.125)
2023 interaction	-0.079
	(0.126)
$\Delta \ln \text{RGDP}_{ct}$	1.403^{***}
	(0.070)
Observations	2,200,635
R^2	0.339

The reference year is 2019. The first row reports its coefficient; year-specific rows report differences from the reference year. Firm–country–product and year fixed effects are included. Standard errors are clustered at the firm level.